

Why Does ETF Short Selling Provide a Different Signal?

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Short selling plays a critical role in determining stock prices and developing efficient markets. The research connecting shorting with return anomalies has even been called one of the great moments in financial economics (Rubinstein [2004]). Short positions are of great interest to investors because they have predictive value for future stock returns.

This return anomaly has two prominent characteristics. First, stocks experiencing high short interest have long-run underperformance (DeChow, Hutton, Meulbroek, and Sloan [2001], Desai, Ramesh, Thiagarajan, and Balachandran [2002], and Boehme, Daniels, and Sorescu [2006]). Second, high-turnover stocks with relatively low short interest have economically significant positive returns (Boehmer, Huszár, and Jordan [2010]).

Popular explanations for this anomaly are based on asymmetric information about stocks (Miller [1977], Hong and Stein [2003], and Diamond and Verrecchia [1987]). Short sellers tend to be sophisticated investors with value-relevant information that other investors do not have. Previous research suggests that this exists for both positive and negative information. In other words, short sellers have demonstrated an ability to profit from overvalued stocks and avoid shorting undervalued stocks.

Does short selling activity have predictive value for securities with minimal asymmetric information? We examine this issue using data on ETFs (exchange-traded funds), which have minimal asymmetric information due to the frequent publication of their net asset values (NAVs). Unlike a regular stock, an ETF can be arbitrated by creating/redeeming shares through the fund sponsor. We use a sample of foreign stock ETFs, which are more likely to have asymmetric information than regular stock ETFs. If short selling activity has predictive value for ETFs, then we are more likely to uncover it in foreign stock ETFs than domestic stock ETFs.

Another justification for examining the short selling of foreign stock ETFs is the useful role it can play in diversification. Increasing correlations among national stock markets create a challenge for international diversification (Christoffersen, Errunza, Jacobs, and Langlois [2012]). Short selling can be particularly valuable in such an environment if countries with low Sharpe ratios can be found (So and Tse [2001]).

One of the easiest ways to achieve a short position in foreign stocks is to sell short ETFs. Miffre [2007] finds that allowing modest 5% short positions in country-specific ETFs improves the mean portfolio return by 17% relative to a long-only portfolio. So and Tse [2001] document that removing the long-

only constraint from international portfolio management increases the Sharpe ratio by 29%.

DATA AND METHODOLOGY

We examine short selling activity in iShares single-country funds, which were introduced in 1996 and target MSCI Country Indexes.¹ Exhibit 1 shows the 21 ETF funds that were trading as of February 2002. Our analysis investigates this group of ETFs because of the long trading history and the similar construction of their benchmark index. Since the iShares country funds were the first foreign stock ETFs to become popular in the U.S., our sample includes 3,419 months of data. The market capitalization of the funds ranges from \$23 million to \$9 billion.

We obtain short sales data from Bloomberg and Capital IQ, and all other market data are sourced from Thomson Datastream. Exhibit 2 describes the closing price, share volume, and short interest over the sample period, which starts with the initial month of trading for each fund and ends with December 2011.

The short interest, which is released monthly, is the number of shares sold short. Relative short interest is short interest divided by the number of shares outstanding. The trading volume shown in Exhibit 2 is the sum of the daily shares traded between monthly short interest releases. On average, investors traded 17 million shares in each of the ETFs per month. Relative short interest averages 8% but varies considerably over the sample period.

Short positions are reported by FINRA member firms as of the settlement on the fifteenth day of each month or the preceding business day. Our analysis computes total returns, which reflect both price changes and dividends paid, between these monthly settlement dates. Assuming annual compounding over the sample period, the average annual ETF return is 5.7%, which is equivalent to a 0.46% monthly return. Abnormal returns equal the ETF return minus the index return.² We use the MSCI World Index and CRSP Value-Weighted Index. From March 1996 to December 2011, the MSCI World Index had a compound annual return of 4.9% while the CRSP Value-Weighted Index returned 6.7%.

EXHIBIT 1

Sample of Foreign Stock ETFs

Symbol	Fund Name	Month of Initial Trading	Market Cap 12/2011 (\$1,000s)
EWA	iShares MSCI Australia Index Fund	3/1996	2,666,202
EWC	iShares MSCI Canada Index Fund	3/1996	4,332,432
EWD	iShares MSCI Sweden Index Fund	3/1996	255,204
EWG	iShares MSCI Germany Index Fund	3/1996	2,278,878
EWH	iShares MSCI Hong Kong Index Fund	3/1996	1,766,700
EWI	iShares MSCI Italy Index Fund	3/1996	98,448
EWJ	iShares MSCI Japan Index Fund	3/1996	5,445,000
EWK	iShares MSCI Belgium Investable Market Index Fund	3/1996	22,946
EWL	iShares MSCI Switzerland Index Fund	3/1996	467,471
EWM	iShares MSCI Malaysia Index Fund	3/1996	838,725
EWN	iShares MSCI Netherlands Investable Market Index Fund	3/1996	74,347
EWO	iShares MSCI Austria Investable Market Index Fund	3/1996	52,650
EWP	iShares MSCI Spain Index Fund	3/1996	101,673
EWQ	iShares MSCI France Index Fund	3/1996	254,184
EWS	iShares MSCI Singapore Index Fund	3/1996	1,323,504
EWU	iShares MSCI Taiwan Index Fund	6/2000	2,213,640
EWV	iShares MSCI United Kingdom Index Fund	3/1996	1,258,488
EWY	iShares MSCI Mexico Investable Index Fund	3/1996	1,107,330
EWZ	iShares MSCI South Korea Index Fund	5/2000	2,990,898
EWZ	iShares MSCI Brazil Index Fund	7/2000	9,238,318
EZA	iShares MSCI South Africa Index Fund	2/2002	479,682

Notes: This exhibit lists the 21 iShares country funds that were trading as of February 2002, when the iShares MSCI South Africa Index Fund was initiated.

EXHIBIT 2

Summary Statistics on ETF Trading

Symbol	Mean Share Price	Mean Volume	Short Interest			Relative Short Interest		
			Min	Max	Mean	Min	Max	Mean
EWA	15.90	26,160,116	2,990	7,268,497	1,238,032	0%	17%	3%
EWC	18.92	18,037,198	0	9,877,763	1,912,098	0%	21%	5%
EWD	21.61	2,427,034	0	930,976	123,038	0%	19%	3%
EWG	20.24	16,726,739	24,728	8,007,215	1,267,550	0%	17%	6%
EWH	12.88	45,824,557	67,800	16,247,794	3,145,098	1%	41%	8%
EWI	21.67	2,446,268	500	6,201,982	482,451	0%	74%	9%
EWJ	10.91	247,227,913	8,000	48,441,979	11,124,905	0%	22%	4%
EWK	15.34	1,921,702	0	1,526,007	207,529	0%	49%	5%
EWL	18.74	2,526,933	41,400	1,584,109	403,135	1%	21%	5%
EWM	8.13	20,377,668	4,750	14,330,348	1,512,248	0%	21%	4%
EWN	20.75	1,147,901	0	1,138,439	108,488	0%	31%	4%
EWO	17.29	2,037,068	575	968,038	173,687	0%	77%	8%
EWP	33.23	2,532,540	23,459	4,143,268	545,290	1%	79%	17%
EWQ	23.22	2,753,123	11,500	3,250,687	364,874	0%	51%	7%
EWS	8.50	27,150,795	12,600	5,464,437	1,114,014	0%	16%	3%
EWT	11.96	120,280,790	58,700	28,118,566	5,614,974	1%	24%	6%
EWU	17.61	10,983,913	1,635	6,473,622	1,049,352	0%	12%	4%
EWV	34.89	36,218,742	426,632	14,979,008	4,721,767	5%	112%	34%
EWY	36.86	34,342,117	37,000	7,633,314	1,930,439	1%	18%	7%
EWZ	39.72	162,713,693	3,000	28,534,639	8,739,043	0%	37%	14%
EZA	75.51	4,848,481	0	2,095,533	696,268	0%	31%	9%

Notes: Trading data are obtained from Thompson Datastream. Short interest data are obtained from Bloomberg and Capital IQ. The volume is the total trading volume between monthly short interest releases. The period for our analysis starts with the initial month of trading for each fund and ends with December 2011.

If short interest activity has predictive value in foreign stock ETFs, we want to know whether it is related to the ability to predict equity returns in foreign markets or currency returns. Equation (1) shows the decomposition of the return on a foreign stock ETF in U.S. dollars (R_s).

$$R_s = R_f + e_i + (R_f)(e_i) \quad (1)$$

R_f is the return on the ETF's portfolio in the foreign market, and e_i is the return on the foreign currency used in the primary market in which the underlying assets trade. Since the return in dollars and the currency return are readily available, we are able to compute the return in the foreign market (R_f). This also allows us to investigate whether prior currency and/or foreign market returns are motivating short sellers to initiate positions.

RESULTS

To analyze the predictive value of short interest activity in foreign stock ETFs, we form deciles based on relative short interest. Each of the deciles contains

about 342 observations on monthly short interest for the ETFs in our sample. Decile 1 contains observations with relative short interest less than 0.7%, while decile 10 contains observations with relative short interest above 16.8%.

For each decile, Exhibit 3 shows the mean abnormal returns for the subsequent month and the two-tailed P-values for the Student's t-test that the mean equals zero. The second column shows that foreign stock ETFs with low levels of short selling activity (decile labeled "low") outperform the MSCI World Index by 0.71% in the following month.³ We reject the hypothesis that the mean abnormal return is zero, since the P-value is 0.03. This is consistent with prior research that indicates highly traded stocks with low short interest have positive excess returns.

Many studies on short selling document that highly shorted stocks experience subsequent underperformance. However, the second column of Exhibit 3 shows a mean abnormal return for high relative short interest ETFs of 0.58% (statistically significant with a

EXHIBIT 3

Mean Abnormal Returns by Short Interest Decile

DECILE	RELATIVE SHORT INTEREST		CHANGE IN RELATIVE SHORT INTEREST	
	Full Sample	Restricted	Full Sample	Restricted
Low	0.71%	1.22%	0.51%	1.14%
	0.03	0.01	0.06	0.09
2	0.24%	0.51%	0.32%	-0.37%
	0.39	0.37	0.29	0.54
3	0.04%	0.04%	0.07%	-0.18%
	0.89	0.93	0.80	0.72
4	0.84%	0.97%	0.10%	0.12%
	0.00	0.05	0.73	0.82
5	0.03%	0.15%	0.41%	0.82%
	0.91	0.77	0.16	0.04
6	0.09%	0.30%	0.12%	0.76%
	0.71	0.56	0.69	0.13
7	0.00%	0.12%	0.26%	0.51%
	1.00	0.83	0.37	0.40
8	0.28%	0.66%	0.04%	0.32%
	0.34	0.18	0.88	0.61
9	0.07%	-0.38%	0.17%	0.55%
	0.82	0.46	0.55	0.30
High	0.58%	0.06%	0.60%	0.20%
	0.08	0.92	0.06	0.68

Notes: Deciles are formed based on either the relative short interest (second and third columns) or the change in relative short interest (last two columns). Abnormal ETF returns are computed using the MSCI World Index. The mean returns for the subsequent month are shown above the two-tailed P-values for Student's t-test that the mean equals zero. The columns that are restricted describe only the observations in which the prior month did not have a change in shares outstanding.

P-value of 0.08). This is a surprising result, as it is not consistent with the reputation of short sellers as sophisticated investors who successfully position themselves in overvalued securities.

To examine the robustness of these results, we compute deciles using the change in relative short interest, and the mean abnormal returns by decile are shown in the fourth column of Exhibit 3. The results are generally the same; both the lowest and highest deciles have positive, statistically significant returns.

ETFs differ from regular stocks because large investors can create and redeem ETF shares through the fund distributor. We explore how this process impacts the predictive ability of short selling activity by calculating the mean returns for only the observations that have no change in the shares outstanding in the prior month. The results are labeled "Restricted" and shown in the third and fifth columns in Exhibit 3. The mean abnormal returns for the high deciles are statistically insignificant from zero for both approaches to forming deciles. Share

creation/redemption in ETFs influences the usual relation between returns and high short interest.

Share Creation/Redemption and Short Selling Activity

To further investigate the effect of share creation/redemption on short selling levels, we categorize our sample based on the change in shares outstanding since the last release of short interest. For each of these categories, Exhibit 4 shows the ETF returns and changes in relative short interest over the same period.

The first row in Exhibit 4 shows the mean returns and change in relative short interest when the ETF shares outstanding decrease by more than 10%. Investors are redeeming shares when the ETF has an average return of -1.64% and underperforms the MSCI World Index by 0.80%. As the mean foreign market portfolio return is five times the size of the mean foreign currency return, the evidence suggests that the poor performance of the

EXHIBIT 4

ETF Returns and Short Interest by Size of Share Creation/Redemption

Change in Shares Outstanding (x)	Mean ETF Return (R_s)	Mean Foreign Market Portfolio Return (R_f)	Mean Foreign Currency Return (e_f)	Mean Abnormal Return	Mean Change in Relative Short Interest	N
$x < -10\%$	-1.64%	-1.42%	-0.26%	-0.80%	2.37%	233
	0.00	0.00	0.25	0.03	0.00	
$-10\% \leq x < -5\%$	-1.29%	-0.69%	-0.63%	-0.14%	0.43%	247
	0.01	0.08	0.00	0.61	0.08	
$-5\% \leq x < 0\%$	-0.08%	-0.12%	0.01%	0.20%	0.15%	485
	0.82	0.70	0.94	0.38	0.25	
$x = 0\%$	0.67%	0.67%	0.03%	0.22%	0.38%	1,114
	0.00	0.00	0.71	0.20	0.00	
$0\% < x < 5\%$	0.71%	0.66%	0.02%	0.02%	-0.09%	493
	0.02	0.01	0.85	0.91	0.59	
$5\% \leq x < 10\%$	1.81%	1.19%	0.59%	0.41%	-0.34%	336
	0.00	0.00	0.00	0.19	0.14	
$x \geq 10\%$	2.82%	2.20%	0.54%	1.38%	-2.21%	523
	0.00	0.00	0.00	0.00	0.00	

Notes: Sample data is categorized based on the change in shares outstanding from the last release of short interest data. The mean returns and change in relative short interest are provided over the same period and shown above the two-tailed P-values for Student's *t*-test that the mean equals zero. The mean ETF return (R_s) is the total return on the ETF shares, and it is decomposed into the foreign market portfolio return (R_f) and foreign currency return (e_f) using Equation (1).

foreign stock market is the primary motivation for the ETF share redemptions.

The mean relative short interest increases 2.37% over this period. This suggests that at the last release of short data, short sellers were avoiding the ETF shares because the shares were priced relatively low compared to their portfolio values in the foreign market. For decreases in shares outstanding between 5% and 10%, the results are largely the same, with share redemptions occurring around statistically significant decreases in ETF returns (-1.29%) and increases in relative short interest (0.43%).

These statistics help us better understand what happens around the redemption of ETF shares. When stock returns in the foreign market are relatively low, investors in the domestic ETF market sell the ETF shares to the point where the price is sufficiently below the NAV to make arbitrage profitable. At this point, the short interest is relatively low, as the ETF shares are undervalued and short sellers avoid them.

Large institutional investors will want to arbitrage the price difference between the ETF shares and the portfolio holdings in the foreign market. They can buy the ETF shares and redeem them for the underlying securities, which can be sold in the foreign market at a relatively high price. As the price difference between the

ETF shares and NAV is minimized, relative short interest increases to a normal level, which is shown as a statistically significant mean increase.

As shown in the bottom row of Exhibit 4, large increases in shares outstanding (at least 10%) occur during months with a mean ETF return of 2.82%. This return is primarily due to the foreign market portfolio return (2.20%) and to a lesser extent the foreign currency return (0.54%). Relative short interest has a statistically significant mean decrease of 2.21% around these large share creations. The smaller increases in shares outstanding offer similar results.

The process of ETF share creation offers an explanation for the decline in relative short interest. When an ETF outperforms the market, investors bid up the price above the NAV. Institutional investors arbitrage the shares by short selling the ETF shares and buying the fund's portfolio in the foreign market. After the shares are created through the fund's distributor, short selling activity declines to a normal level.

Time Series Analysis

We develop a time-series model of relative short interest in ETFs to go beyond the basic descriptive statistics already provided. The distribution of the rela-

EXHIBIT 5

Model of Relative Short Interest

Independent Variable	Estimate	Standard Error	t-statistic	P-value
Intercept	-3.56	0.08	-43.49	0.00
Volume	0.40	0.05	8.76	0.00
SO Change	0.48	0.07	7.35	0.00
AR1	-0.6	0.02	-36.06	0.00
AR2	-0.1	0.02	-5.40	0.00
AR3	-0.1	0.02	-5.84	0.00

Notes: This exhibit shows regression analysis results for a time-series model of the natural log of relative short interest. The sample includes 3,418 months of short interest data. Volume is the number of shares traded since the last short interest release divided by the number of shares outstanding. SO Change is the change in shares outstanding that occurs before the next release of short interest data. The P-value is for two-sided test of significance.

tive short interest is normalized by taking the natural log. Since significant positive autocorrelation exists, we employ the Ljung-Box test to determine the best model, which is an AR(3) model.

Exhibit 5 shows the model estimates, which include two independent variables. Since short selling activity is affected by liquidity, the first independent variable is volume, which equals the shares traded since the last short interest release divided by the shares outstanding. As shown by the estimated coefficient of 0.4, relative short interest increases with ETF liquidity, which makes short selling less costly.

The second independent variable is labeled SO Change. It equals the change in shares outstanding that occurs before the next release of short interest data. The estimated coefficient is 0.48 and consistent with the prior explanation that short interest is relatively high before an increase in shares outstanding through the creation of ETF shares. The regression also includes three terms to account for the negative serial correlation in short interest. The regression R^2 , which measures the predictability of the explanatory variables aside from the autocorrelation, equals 66% for this model.

Analysis of Unexpected Short Selling

Exhibit 5 presents a model of short interest, and the residuals from this model represent a measure of the unexpected level of short selling activity. To better under-

EXHIBIT 6

Model of Unexpected Relative Short Interest

Independent Variable	Estimate	Standard Error	t-statistic	P-value
Intercept	0.01	0.001	12.58	0.00
$R_{p,t}$ lag 1 month	0.06	0.013	4.15	0.00
$R_{p,t}$ lag 2 months	0.01	0.013	0.59	0.56
$R_{p,t}$ lag 3 months	-0.02	0.013	-1.14	0.25
$e_{i,t}$ lag 1 month	0.06	0.030	1.92	0.06
$e_{i,t}$ lag 2 months	-0.02	0.030	-0.59	0.56
$e_{i,t}$ lag 3 months	-0.04	0.029	-1.36	0.18

Notes: This table shows regression analysis results for a time-series model of the unexpected level of relative short interest, which is defined as the residuals from the model presented in Exhibit 5. $R_{p,t}$ is the return on the ETF's portfolio in the foreign market. $e_{i,t}$ is the return on the foreign currency used in the primary market in which the underlying assets trade. $R_{p,t}$ and $e_{i,t}$ are included in the model as lagged variables for each of the prior three months. The sample includes 3,418 months of short interest data. The P-value is for two-sided test of significance.

stand the drivers of short positions in ETFs, we examine the relation between past returns and unexpected short interest. Exhibit 6 shows the estimates from a model of unexpected relative short interest with six measures of past returns. The first three independent variables in this model are the ETF's portfolio returns in the foreign market ($R_{p,t}$) for each of the prior three months. The model also includes the foreign currency return ($e_{i,t}$) for each of the past three months.

Unexpected relative short interest in ETFs has a positive, statistically significant relation to the prior month's foreign market return and exchange rate return (both coefficient estimates are 0.06). We fail to reject the hypothesis that these coefficients are equal, and the test has a P-value of 0.96. None of the other prior returns are statistically significant. Therefore, ETF short sellers appear to be contrarian, but they are not influenced more by foreign market returns than currency returns. The R^2 of the regression is 0.01, which indicates that past returns are not an economically significant driver of ETF short selling.

Next, we relate the unexpected relative short interest to the ETF return in the subsequent month. We form deciles using unexpected relative short interest, and Exhibit 7 shows the mean abnormal ETF return for the subsequent month. Abnormal returns are computed using the MSCI World Index and the CRSP Value-Weighted Index. The P-values for a two-sided test that

EXHIBIT 7

Mean Forward Returns by Unexpected Short Interest Decile

DECILE	Mean Abnormal Return, MSCI	Mean Abnormal Return, CRSP
Low	0.67%	0.71%
	0.02	0.02
2	0.31%	0.40%
	0.26	0.16
3	0.68%	0.79%
	0.01	0.00
4	0.02%	0.11%
	0.95	0.71
5	0.42%	0.17%
	0.17	0.56
6	0.46%	0.32%
	0.08	0.24
7	0.21%	0.27%
	0.49	0.34
8	-0.12%	-0.34%
	0.69	0.23
9	-0.28%	-0.52%
	0.37	0.09
High	0.29%	0.22%
	0.35	0.48

Notes: Deciles are formed based on the unexpected level of relative short interest. Mean abnormal ETF returns for the subsequent month are shown above the two-tailed P-values for Student's t-test that the mean equals zero. The middle column computes the abnormal return using the MSCI World Index, and the column to the right uses the CRSP Value-Weighted Index. The statistics are calculated using 3,454 months of short interest data.

the mean equals zero are shown below the mean returns. The decile associated with low unexpected short selling activity has significant positive abnormal returns under both approaches for computing abnormal returns.

The third decile also shows significant positive returns. None of the other deciles show statistically significant results under both approaches. Consistent with the initial results presented in Exhibit 3, the statistically significant ETF returns associated with high short interest are not observed when adjusting for share arbitrage and liquidity.

What insights do short sellers have on ETFs in the low short interest decile? The mean foreign market portfolio return (R_f) for this decile is 1.28%, which is statistically different from zero with a P-value = 0.00. The mean foreign currency return (e_f) is 0.08% and it is not statistically significant (P-value = 0.64). Short sellers are avoiding the foreign stock ETFs in the month before these foreign stock markets experience high returns.

CONCLUSIONS

Short selling is of great interest to investors because this activity has predictive value for future stock returns. Stocks with high short interest underperform, and high volume stocks with low short interest outperform. Why don't stock prices fully reflect this readily available information? One theory is that there is asymmetric information about stocks, and short sellers tend to be sophisticated investors with superior insight into a stock's prospects.

We examine a sample of ETFs, which are securities with easily observable values and thus minimal asymmetric information. In contrast to regular stocks, ETFs with high short interest experience positive abnormal returns. Our analysis suggests that the creation and redemption of ETF shares influence the level of short selling. Investors who casually apply the Wall Street lore about short selling to ETFs do so at their peril.

Foreign stock ETFs with low short interest have positive abnormal returns. This finding is robust to changes in shares outstanding, and whether one examines unadjusted short interest or unexpected short interest. These abnormal returns are typically caused by higher prices in foreign stock markets and not exchange rate changes. Boehmer, Huszár, and Jordan [2010] also find excess returns for regular stocks. Our study provides an important extension to this line of research by showing that it exists for securities with minimal asymmetric information.

ENDNOTES

¹These funds were originally called WEBS. The MSCI Country Indexes are value weighted and represent about 60% of each country's stock market capitalization.

²We use this approach to computing abnormal returns because these ETFs are diversified across stocks and industries.

³While not presented in Exhibit 3, results using the CRSP Value-Weighted Index are similar.

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